

$$= 33 \times (20 \times \pi \times 0.33) = 170 \text{ W}$$

7

Ans: A

Assume that the horizontal force at the top hinge is to the right,

Taking moment about bottom hinge,

$$(T \sin \theta)(d) + (T \cos \theta)(d) = (R_x)(d) + (W)(0.5d)$$

$$(150 \sin 30^\circ)(0.5) + (150 \cos 30^\circ)(0.5) = R_x(0.5) + (20)(9.81)(0.5)(0.5)$$

$$R_x = 107 \text{ N to the right. (since the answer is positive)}$$

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Ans : C